

# IEEE 802.15.4

- **Small packet** size (the maximum transmission unit or MTU on IEEE 802.15.4 links is **127 bytes**), which provides even **less room** for data when including other header
- Support for both 16-bit short or IEEE 64-bit extended media access control (MAC) addresses.
- **Low data rates**; the IEEE 802.15.4 specification allows various data rates from 20Kbits/s (868 MHz) to 250Kbits/s (2.45 GHz).
- Support of star and mesh topologies.

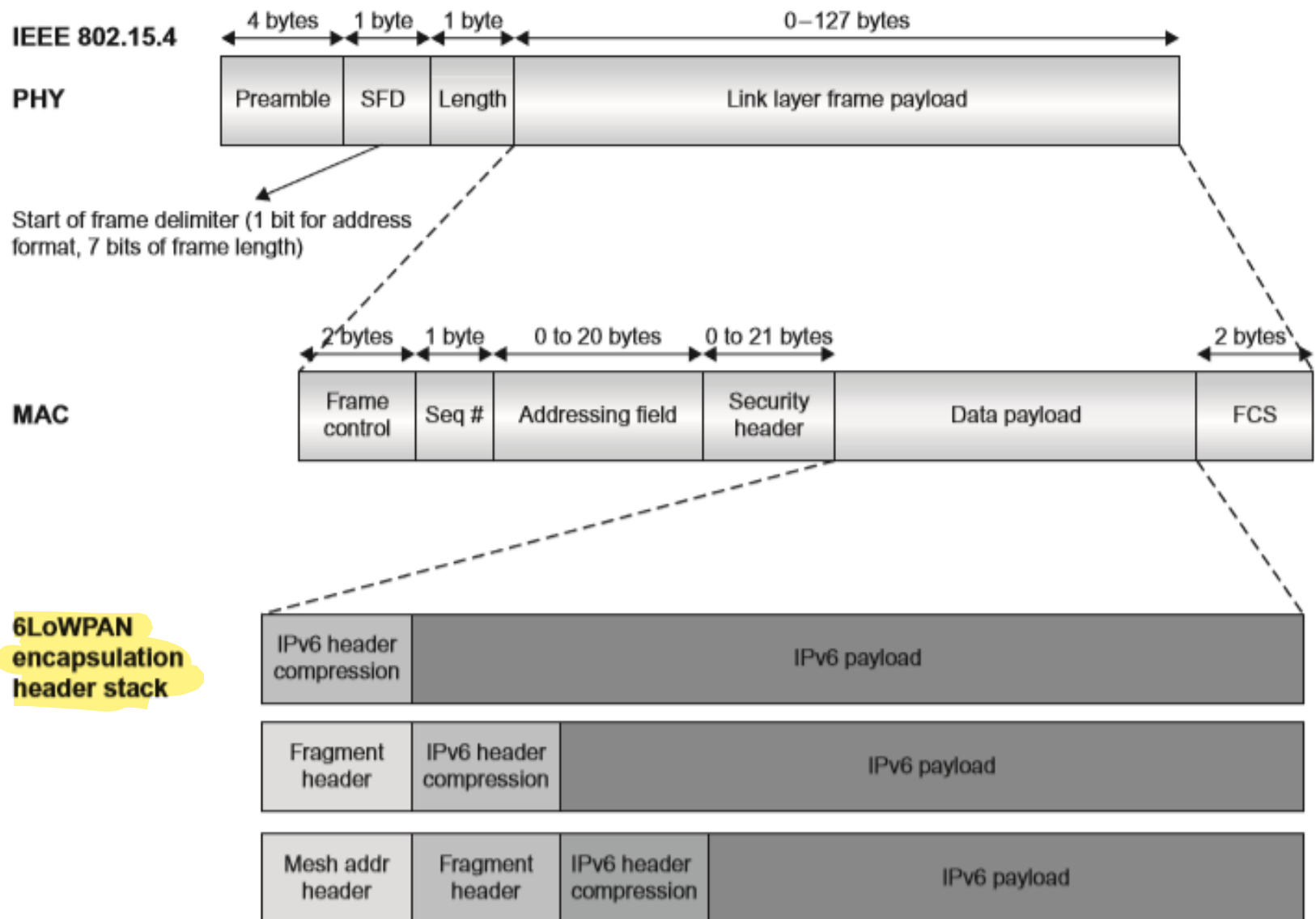
# THE 6LoWPAN ADAPTATION LAYER

- Since **IPv6 mandates** supporting links with an MTU (Maximum Transmission Unit) of **1280 bytes**, it was necessary for IEEE 802.15.4 links that have an MTU of 127 bytes to specify an adaptation layer below IP responsible for handling packet fragmentation and reassembly.
- The MTU size of IEEE 802.15.4 links was **purposely small** to cope with limited **buffering capabilities** and to limit the packet error rate since the **bit error rate (BER)** is relatively high.
- an adaptation layer is needed to comply with the IPv6 requirement to support a minimum MTU size of 1280 bytes as well as to support **compression techniques** to reduce protocol overhead.

# The 6LoWPAN adaptation layer provides three main services:

- Packet fragmentation and reassembly
- Header compression
- Link layer (layer 2) forwarding when multi-hop is used by the link layer

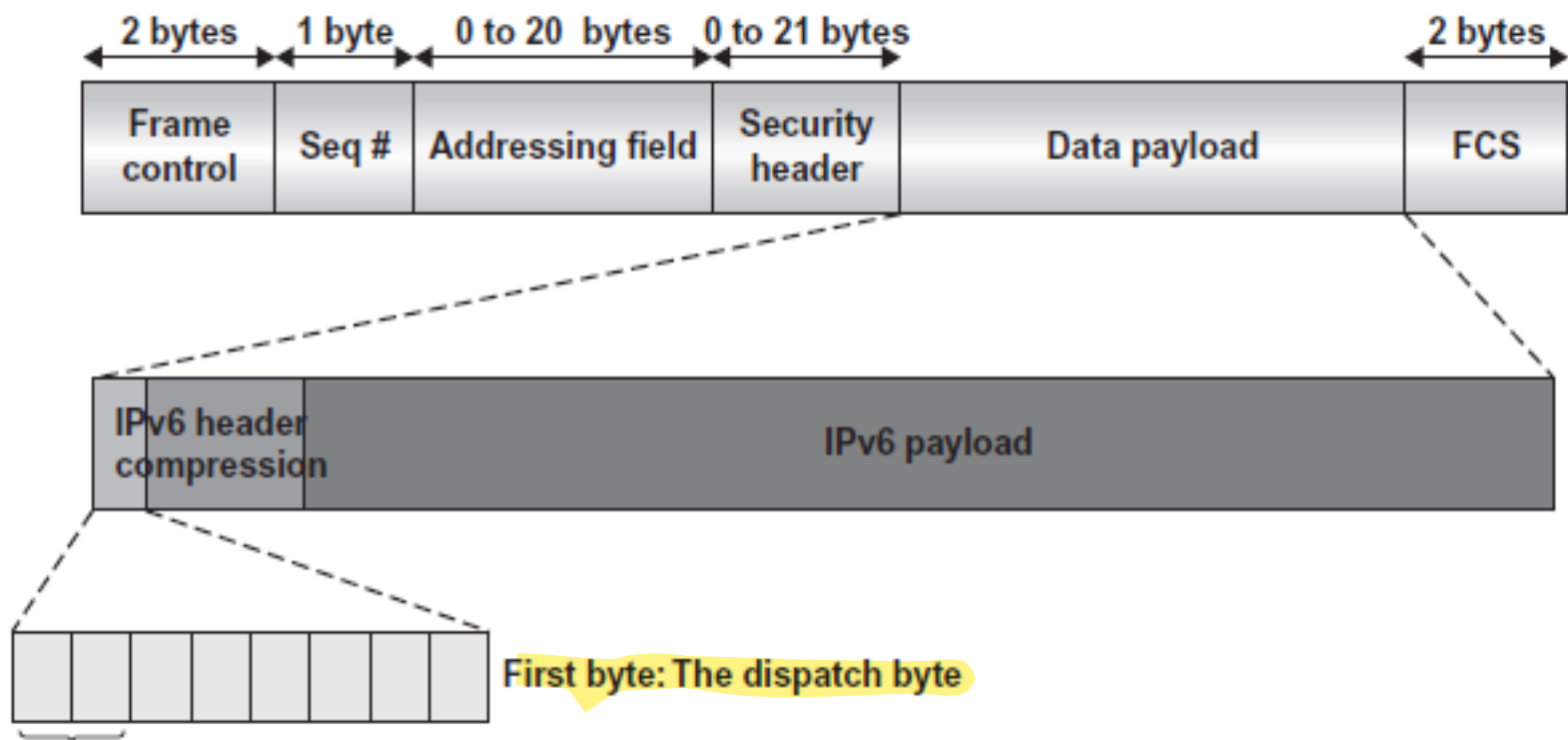
- The 6LoWPAN adaptation currently supports three headers:
  - a mesh addressing header,
  - the fragment header, and
  - the IPv6 header compression header(they must appear in that order when present).



**FIGURE 16.1**

6LoWPAN encapsulation header stack.

# The 6LoWPAN dispatch byte (first byte)



|    |                                                |
|----|------------------------------------------------|
| 00 | Not a 6LoWPAN frame                            |
| 01 | IPv6 addressing header                         |
| 10 | Mesh header                                    |
| 11 | Fragmentation header (6 lower bits are 100xxx) |

**FIGURE 16.2**

Dispatch byte of the IPv6 header compression header.

## The 6LoWPAN dispatch byte (first byte)



First byte: The dispatch byte

| Pattern   | Header type                               |
|-----------|-------------------------------------------|
| 00 xxxxxx | NALP - not a LoWPAN frame                 |
| 01 000001 | IPv6 - uncompressed IPv6 addresses        |
| 01 000010 | LOWPAN_HC1-LOWPAN_HC1 compressed IPv6     |
| 01 000011 | reserved - reserved for future use        |
| ...       | reserved - reserved for future use        |
| 01 001111 | reserved - reserved for future use        |
| 01 010000 | LOWPAN_BCO - LOWPAN_BCO broadcast         |
| 01 010001 | reserved - reserved for future use        |
| ...       | reserved - reserved for future use        |
| 01 111110 | reserved - reserved for future use        |
| 01 111111 | ESC - additional dispatch byte follows    |
| 10 xxxxxx | MESH - Mesh header                        |
| 11 000xxx | FRAG1 - fragmentation header (first)      |
| 11 001000 | reserved - reserved for future use        |
| ...       | reserved - reserved for future use        |
| 11 011111 | reserved - reserved for future use        |
| 11 100xxx | FRAGN - fragmentation header (subsequent) |
| 11 101000 | reserved - reserved for future use        |
| ...       | reserved - reserved for future use        |
| 11 111111 | reserved - reserved for future use        |

**FIGURE 16.3**

Value of the 6LoWPAN dispatch byte.

- The first byte of the encapsulation header identifies the next header.
- For example, if the first 2 bits are equal to 11, the next header is a **fragmentation** header.
- If the first 8 bits are equal to 01000001, what follows is an IPv6 *uncompressed packet*. *In contrast*, a value of 01000010 indicates that what follows is a header related to a compressed header using HC1 compression

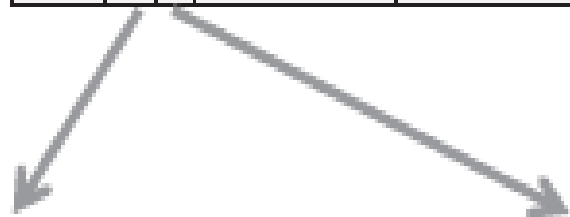
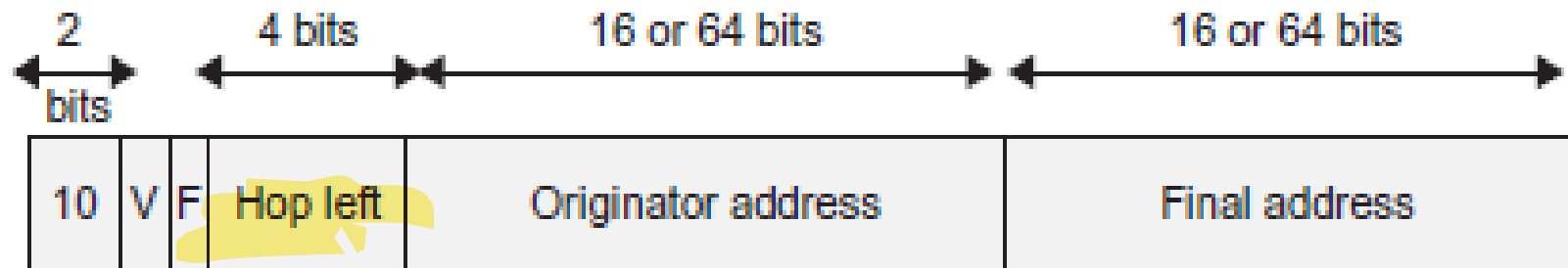


# The Mesh Addressing Header

- The mesh addressing header is used in conjunction with a mesh-under “routing ” approach where nodes that are **not in direct communication** make use of **multi-hop** “ routing ” at the link layer using link layer addresses.
- According to IEEE 802.15.4, only **full function devices (FFDs)** perform mesh-under operation. **Reduced function devices (RFDs)** systematically send all of their traffic to FFDs.

- **The source and destination nodes** are then referred to as the originator and final destination, respectively.
- the first 2 bits of the dispatch byte identify the presence of a mesh header and are equal to 10.
- **Bit 2 (V, Very first bit):**
- 0: The originator address is an IEEE extended 64-bit address (EUI-64).
- 1: The originator address is a short 16-bit address.
- **Bit 3 (F, Final destination):**
- 0: The final address is an IEEE extended 64-bit address (EUI-64).
- 1: The final address is a short 16-bit address.

## Mesh header



V=0 originator 64-bit  
EUI address  
V=1 originator 16-bit  
short address

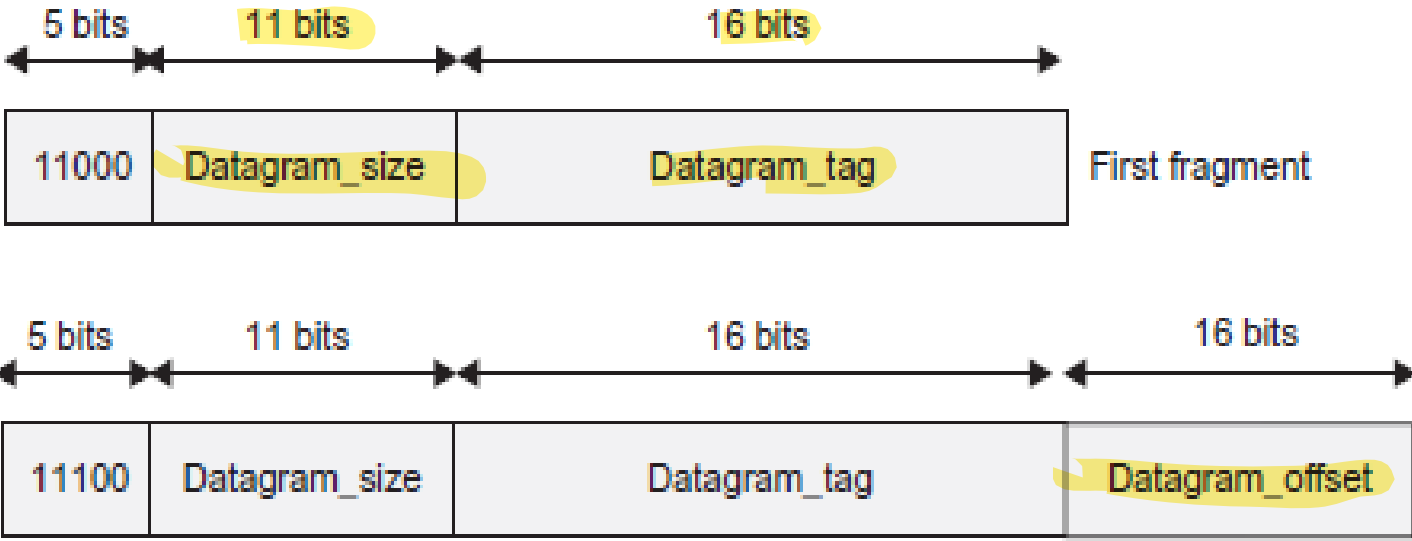
V=0 final destination  
64-bit EUI address  
V=1 final destination  
16-bit short address

- **Bits 4 through 7 (HopLeft):**
- The HopLeft field value is decremented by each node before sending the packet to its next hop.
- When the HopLeft field reaches the value of 0, the packet is simply discarded.
- When equal to 15, an additional byte (called the deep hops left) immediately follows when forwarding along a path with more than 14 hops is needed.
- The originator and final link layer address fields then follow (16 or 64 bits).

- Thus the set of link layer addresses is as follows. When a node A sends a frame to a final destination C via the node B:
- The **originator address** of the mesh header is set to the link layer address of A.
- The **final destination address** of the mesh header is set to the link layer address of C.
- The **source address of the IEEE 802.15.4** frame is the address of the node sending the frame (A).
- **The destination address of the IEEE 802.15.4** frame is the link layer address of the next-hop node as determined by the mesh-under routing protocol (B in this example).

- Upon receiving the frame, B performs the following process:
- The hop left field is decremented.
- If the hop left field is not equal to 0 (if equal to 0, the frame is discarded), then B determines that the next hop is C.
- The originator and final destination address of the mesh header are unchanged.
- The source address of the IEEE 802.15.4 frame is set to the link layer address of B.
- The destination address of the IEEE 802.15.4 frame is set to the link layer address of C.

Fragment header



# 6LoWPAN Header Compression

- Header Compression Using LOWPAN\_HC1 focused on highly optimizing the compression unicast link-local addresses
- Encoding technique (IPHC) –multicast addresses
- LOWPAN\_HC2-The HC\_UDP Compression Technique



The HC1 Compression Technique The HC1 compression technique relies on the following observations:

- IP version is always 6.
- Since HC1 is optimized for link-local addresses, the IPv6 interface ID (bottom 64 bits of the IPv6 address) can be inferred from the link layer MAC address.
- The packet length can be inferred from the frame length field of the IEEE 802.15.4 frame (or from the datagram size field of the fragment header when present).
- Common value for the TC and flow label is 0
- Next header is UDP, TCP, or ICM

